## Normal operator

**Lemma.** Let $T: V \to V$ be a linear operator on fin. dim inner product space $V$.
If $\lambda$ is an eigenvalue of $T$, then $\bar{\lambda}$ is an eigenvalue of $T^*$.

**Proof.** Let $v$ be an eigenvector of $\lambda$. Then, for any $x \in V$,

$$0 = \langle 0, x \rangle = \langle (T - \lambda I)(v), x \rangle = \langle v, (T - \lambda I)^*(x) \rangle$$

That is, $v \in \left(\mathrm{Im}(T^* - \bar{\lambda} I)\right)^{\perp} = \ker(T^* - \bar{\lambda} I) \neq \{0\}$.

## Schur's Lemma

Let $T: V \to V$ be a linear operator on fin. dim inner product space $V$ over a field $F$.
If a characteristic polynomial $f(t)$ of $T$ splits over $F$,
then there exists an orthonormal basis $\gamma$ for $V$ s.t. $[T]_\gamma$ is upper-triangle matrix.

Def. Let $T: V \to V$ be a linear operator on inner product space $V$.
If $T^* \circ T = T \circ T^*$, then $T$ is called Normal operator

Proposition, Let $T: V \to V$ be a Normal operator on inner product space $V$.

1) For any $x \in V$, $\|T(x)\| = \|T^*(x)\|$.
2) For any $c \in F$, $T - cI$ is Normal.
3) For any $\lambda \in F$ and $x \in V$, $T(x) = \lambda x \iff T^*(x) = \bar{\lambda} x$
4). For any distinct eigenvalue $\lambda_1, \lambda_2$ of $T$, $x \in E_{\lambda_1}$ and $y \in E_{\lambda_2} \Rightarrow \langle x, y \rangle = 0$.

「Hermition

Let $V$ be an Inner product space.

⌈Prop. Let $T, U : V \to V$ be Hermitian.

$$T \circ U \text{ is Hermitian iff } T \circ U = U \circ T$$

Prof. $T \circ U$ is Hermitian $\Leftrightarrow T \circ U = (T \circ U)^* = U^* \circ T^* = U \circ T$.

⌈Prop. Let $T : V \to V$ be a linear operator, and $W \leq V$ be a $T$-invariant subspace

1. If $T$ is Hermitian, then a restriction $T|_W$ is Hermitian.
2. $W^\perp$ is $T^*$-invariant subspace
3. If $W$ is $T, T^*$-invariant, $T|_W^* = T^*|_W$.
4. If $W$ is $T, T^*$-invariant and $T$ is Normal, then $T|_W$ is Normal.

Prof. 1) is clear; for any $x, y \in W$;

$$\langle T|_W(x), y \rangle = \langle T(x), y \rangle = \langle x, T^*(y) \rangle = \langle x, T(y) \rangle = \langle x, T|_W(y) \rangle$$

2). Let $x \in W^\perp$ be given. Then, for $y \in W$,

$\langle T^*(x), y \rangle = \langle x, T(y) \rangle = 0$, because $T(y) \in W$. Thus $T^*(x) \in W^\perp$.

3). Let $x, y \in W$ be given. Then,

$$\langle T(x), y \rangle \overset{①}{\underset{②}{=}} \langle T|_W(x), y \rangle = \langle x, T|_W^*(y) \rangle$$
$$= \langle x, T^*(y) \rangle = \langle x, T^*|_W(y) \rangle.$$

That is, for any $x, y \in W$,

$$\langle x, T|_W^*(y) - T^*|_W(y) \rangle = 0.$$

This implies $T|_W^* = T^*|_W$

4) Directly, given by 3.

⌈Prop. Let $V$ be a fin. dim. Complex. inn. prod. space, let $T : V \to V$ be a Normal.

If $W \leq V$ is $T$-invariant, then also $T^*$-invariant.

Prof. Since $T$ is Normal operator on fin. dim. Comp. inn space, $T$ is diagonalizable by an orthonormal $\beta$.

Let $x \in W$ be given with a representation $\sum a_i v_i$ where $v_i \in \beta$. Then,

$T^*(x) = \sum a_i T^*(v_i) = \sum a_i \bar{\lambda}_i v_i \in W$, since $v_i \in W$.

Prop. If $T: V \to V$ is Normal, then

$$T(x)=0 \iff \|T(x)\|=0 \iff \|T^*(x)\|=0 \iff T^*(x)=0.$$

Thus, $\ker T = \ker T^*$.

Moreover, if $V$ is fin. dim, then

$$\operatorname{Im} T = \operatorname{Im} T^*$$

Thm. Let $V$ be a fin. dim, inn. space, $T: V \to V$ be a Hermitian.

For any $x \in V$,

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2.$$

Therefore, $T-iI$ is Invertible and $((T-iI)^{-1})^* = (T+iI)^{-1}$.

Prof. By lemma,

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2 \pm 2\operatorname{Re}\langle T(x), ix\rangle$$

Meanwhile, for any $x \in V$,

$$\langle T(x), ix\rangle = -i\langle T(x), x\rangle$$

$$\langle T^*(x), ix\rangle = \langle x, iT(x)\rangle = -i\langle x, T(x)\rangle$$

This implies $\operatorname{Re}\langle T(x), ix\rangle = 0$.

Now,

$$\begin{aligned}\ker(T \pm iI) &= \{x \in V \mid (T \pm iI)(x) = 0\} \\ &= \{x \in V \mid \|T(x)\|^2 + \|x\|^2 = 0\} \\ &= \{0\}.\end{aligned}$$

Prop. Let $V$ be a Complex. inn. prod space.

1) $T$ is Hermitian $\Leftrightarrow$ for all $x \in V$, $\langle T(x), x\rangle \in \mathbb{R}$.

2). If for all $x \in V$, $\langle T(x), x\rangle = 0$, then $T = T_0$.

Proof. At first, show that 2). For any $x, y \in V$,

$$0 = \langle T(x+y), x+y\rangle = \langle T(x), x\rangle + \langle T(x), y\rangle + \langle T(y), x\rangle + \langle T(y), x\rangle = \langle T(x), y\rangle + \langle T(y), x\rangle$$

$$0 = \langle T(x+iy), x+iy\rangle = \langle T(x), x\rangle + \langle T(x), iy\rangle + \langle T(iy), x\rangle + \langle T(iy), iy\rangle = \langle T(x), iy\rangle + \langle T(iy), x\rangle$$

Now, $\langle T(x), y\rangle + \langle T(y), x\rangle = 0$ and $-\langle T(x), y\rangle + \langle T(y), x\rangle = 0$.

That is, $\langle T(x), y\rangle = 0$. Since $x, y$ are chosen arbitrarily, $T = T_0$.

Suppose $T$ is Hermitian. Then, for any $x \in V$,

$$\langle T(x), x\rangle = \langle x, T^*(x)\rangle = \langle x, T(x)\rangle = \overline{\langle T(x), x\rangle}$$

Thus $\langle T(x), x\rangle \in \mathbb{R}$.

Conversely, Suppose for all $x \in V$, $\langle T(x), x\rangle \in \mathbb{R}$. Then, directly $\langle T^*(x), x\rangle \in \mathbb{R}$.

Now, 
$$\begin{aligned}\langle (T - T^*)(x), x\rangle &= \langle T(x), x\rangle - \langle T^*(x), x\rangle \\ &= \langle T(x), x\rangle - \langle x, T(x)\rangle \\ &= 0.\end{aligned}$$

By 2), we obtain the result.

Prop. $A = B^t B$ for some $B \in M_n(\mathbb{R})$, $A$ is called Gramian matrix.

$A$ is Gramian if and only if $A^t = A$ and $A$ has no negative eigenvalue.

Proof. Suppose that $A$ is Gramian. Then, clearly symmetric.

Since $L_A = L_{B^tB} = (L_{B^t}) \circ (L_B)$,

$$\langle L_A(x), x\rangle = \langle L_B(x), L_B(x)\rangle \geq 0.$$

Conversely, For each $1 \leq i, j \leq n$,

$$([U^*]_\beta)_{ij} = \langle U^*(v_j), v_i\rangle = \langle v_j, U(v_i)\rangle = \sqrt{\lambda_i} \cdot \langle v_j, v_i\rangle = \sqrt{\lambda_i}\, \delta_{ij}$$

Meanwhile, $T(v_i) = (U^* \circ U)(v_i)$ $\quad [T]_\beta = A = [U]_\beta^* [U]_\beta$.

Def. Let $V$ be a fin. dim. inn. space, $T: V \to V$ be a Hermitian

$T$ is called positive semidefinite if

$$\forall x \in V, \ \langle T(x), x \rangle \geq 0$$

In addition if $\langle T(x), x \rangle = 0 \iff x = 0$, then $T$ is called positive definite.

Thm. Let $V$ be an inn. space with $\dim V = n$, and $\beta$ be an orthonormal basis for $V$.

Let $T: V \to V$ be Hermitian, $A = [T]_\beta$.

1). $T$ is positive semidefinite if and only if every eigenvalue of $T$ is non-negative.

2) $T$ is positive definite if and only if every eigenvalue of $T$ is positive.

proof. Let $\gamma$ be an orthonormal basis consisted by eigenvectors of $T$.

Let $x \in V$ be given. Denote $\gamma = \{v_1, \cdots, v_n\}$, $B = [T]_\gamma$ Then,

$$x = \sum_{i=1}^n a_i v_i \quad \text{for some } a_i \in F.$$

Now,
$$\begin{aligned}
\langle T(x), x \rangle &= \langle T(\sum_{i=1}^n a_i v_i), \sum_{i=1}^n a_i v_i \rangle \\
&= \langle \sum_{i=1}^n a_i \lambda_i v_i, \sum_{i=1}^n a_i v_i \rangle \\
&= \lambda_i \cdot \langle \sum_{i=1}^n a_i v_i, \sum_{i=1}^n a_i v_i \rangle \\
&= \sum_{i=1}^n \lambda_i \cdot \|a_i v_i\|^2
\end{aligned}$$

3). $T$ is positive definite if and only if for any $(a_1, \cdots, a_n) \in F^n$, $\sum_{i,j} A_{ij} a_j \overline{a_i} > 0$.

proof. put $\beta = \{w_1, \cdots, w_n\}$. Then, for $\underline{\sum a_i w_i}$,

$$\begin{aligned}
&\langle T(\sum a_i w_i), \sum a_j w_j \rangle \\
&= \sum_{i,j} a_i \overline{a_j} \underline{\langle T(w_i), w_j \rangle} \\
&= \sum \underline{A_{ji}} a_i \overline{a_j}
\end{aligned}$$

Prop. If $T^2 = U^2$ and $T, U$ are positive semidefinite, then $T = U$.

Proof, Let $\beta = \{v_1, \cdots, v_n\}$ be an orthonormal consisted eigen- $T$.

For each $1 \leq i \leq n$, $T^2(v_i) = \lambda_i^2 v_i = U^2(v_i)$.

If $\lambda_i = 0$, then cleary $T(v_i) = 0$, and

$$0 = \langle U^2(v_i), v_i \rangle = \langle U(v_i), U(v_i) \rangle$$

Thus $U(v_i) = 0$.

Now, assume $\lambda_i > 0$, $\therefore$

$$0 = (U^2 - \lambda_i^2 I)(v_i) = (U + \lambda_i I) \circ (U - \lambda_i I)(v_i)$$

Since $U$ is positive-semidefinite, every eigenvalue is bigger than zero.

Thus $U + \lambda_i I$ is invertible. Now, $(U - \lambda_i I)(v_i) = 0$. $\therefore U(v_i) = \lambda_i v_i = T(v_i)$.